The Perils of Bilateral Sovereign Debt*

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Abstract

We study the interaction between private and official sovereign debts. We develop a quantitative sovereign default model featuring a senior lender with whom borrowing terms are negotiated. We use this model to evaluate implications of the emergence of new official creditors not bound by the Paris Club framework. The dynamics of bilateral bargaining lead the government to issue more market debt, raising default risk and creating welfare losses. This relational overborrowing effect arises due to an endogenous cross-elasticity of bilateral terms to market debt, which can be assessed in practice to evaluate new forms of bilateral sovereign debt.

JEL Classification F34, F41, G15
Keywords Sovereign debt, debt dilution, bilateral bargaining, official debt

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Introduction

A large fraction of sovereign borrowing in emerging-market economies takes the form of official debt, including loans from other governments, regional development banks, or multilateral institutions such as the IMF and the World Bank. The past few decades have witnessed the rise of new sovereign bilateral creditors operating outside of the Paris Club institutional framework (Horn et al., 2021b; Gelpern et al., 2021). The claims to de facto senior creditor status (relative to private creditors) by these new lenders have raised concerns about welfare implications for borrowing countries (Horn et al., 2021a, 2023).

This paper evaluates such concerns in the context of a quantitative sovereign debt and default model, augmented by the presence of a large lender who offers an alternative source of funds. Relative to private creditors in capital markets, this large lender has a superior enforcement technology. This assumption allows us to model the potential success of the new sovereign creditors in becoming senior to bondholders in international capital markets and trace out its implications. In addition, in our baseline setup, we assume that borrowing terms with the large lender are determined through bargaining, which gives a crucial role to the dynamics of the bilateral relationship.

We focus on the interaction between private markets and the large lender as two potential sources of financing.[1] While the availability of bilateral loans from the large lender affects the government's behavior in private debt markets, outcomes in these markets also influence threat points in the bilateral negotiation. The interest rate charged by the large lender is therefore constrained by implicit competition from private debt markets. But when default risk pushes up yields on marketable debt, the large lender is also able to charge a premium. Because there is no default risk on the bilateral loan, such a premium only reflects the borrower's outside options.

The baseline model yields two main results: the presence of the senior lender leads to a form of overborrowing in debt markets, and this creates welfare losses for the government, relative to a benchmark in which only market funding is available. While bilateral loans can and are often used on the equilibrium path to avoid costly defaults, the borrowing country is worse off when the large lender is present. One reason for this is that the possibility of borrowing from the large lender while being excluded from private markets raises the value of default, which, in turn, increases default incentives and lowers (marketable) debt prices. But there is another, more fundamental reason for the welfare losses. Even imposing the constraint that bilateral financing cannot be used during default, welfare for the government is still reduced by the presence of the large lender due to a "relational overborrowing" effect. Financing terms with the large lender

[1] As most of the quantitative literature on sovereign debt, we focus on debt as a vehicle for consumption smoothing or frontloading and abstract from project financing or development lending.

are determined by both parties' threat points, and the government understands that reaching the bargaining stage with little cash on hand weakens its bargaining position. This effect is captured and summarized by an endogenous cross-elasticity of bilateral loan terms to market debt issuances. The cross-elasticity acts as a countervailing force for the market discipline of spreads and generates an extra incentive for the government to issue debt more aggressively and delever more slowly. The end result is an increase in the default frequency and therefore in spreads, which ultimately create welfare losses for the government.

The dynamics and welfare effects we describe result from three critical assumptions regarding the large creditor and the funding it offers. Compared to competitive markets, we assume that it is more difficult (or impossible) to default on the large creditor, that its loans are of shorter maturity, and that the terms of borrowing result from bilateral bargaining rather than competition and a zero-profit condition. Central Bank swap lines constitute a prime real-world example, which Horn et al. (2021b) identify as a dominant channel through which official bilateral lending has reemerged since the 2000s.[2] These facilities are typically short-term and feature none of the cross-default clauses present in government bonds, facilitating seniority. They also involve no recorded defaults. Finally, while their terms are typically strictly confidential, anecdotal evidence suggests that a fair amount of bilateral negotiation is involved in determining amounts and interest rates, rather than market mechanisms.

Other forms of official debt share the seniority and/or duration features emphasized in our baseline model—including, for example, some types of IMF programs and repo or swap facilities offered by major central banks such as the Fed or the ECB. However, unlike the bilateral bargaining framework we assume, these facilities typically feature borrowing terms that are fixed in advance. For instance, IMF lending is subject to surcharges based on access and duration thresholds (IMF, 2024b), while the Fed's swap lines carry a standard 25 basis point spread over policy rates (Bahaj and Reis, 2023) and are publicly reported in real time on the FRBNY's website. To accommodate such institutional designs, we develop an extension in which bilateral loan pricing does not result from bargaining but rather follows fixed rules. This allows us to explore how rules-based lending affects debt dynamics and to compare welfare outcomes across alternative institutional arrangements.

The extension shows, in a much simpler setup, that when the interest rate on bilateral loans is relatively insensitive to the government's market debt—that is, when the interest rate is fixed or has a low elasticity to market debt—the relational overborrowing channel is muted or disappears altogether. In this case, the dominant effect of the large lender is to help the government

[2]While swap lines are important in our motivation to study the interaction of market debt and (non-defaultable) bilateral loans, our model is not geared to represent all aspects of swap lines. For example, we do not include a difference in the currency denomination of bilateral and market debt, conditionality (including requirements on the level of foreign reserves), or collateral requirements.

avoid costly defaults on its marketable debt. As a result, the equilibrium features a lower default frequency and reduced sovereign spreads, improving the welfare of the borrowing country. We also find that an optimal design of bilateral lending facilities involves the *opposite* cross-elasticity: it makes the bilateral interest rate rise with the government's net market financing. This design, which is optimal within a parametric family of rules, reinforces the disciplining role of spreads and delivers higher welfare for the government relative to the equilibrium with bargaining and also relative to the equilibrium with only market funding.

A key parameter in the model is the government's bargaining power relative to the large lender. When the government can make take-it-or-leave-it offers (i.e., when it has full bargaining power), the large lender simply provides non-defaultable loans at the risk-free rate, recovering the model of Hatchondo, Martinez, and Önder (2017). In this case, the government's welfare is also higher relative to the equilibrium without the large lender. In our calibration, however, such gains quickly dissipate as the government's bargaining power declines.

In the baseline model, we assume that the large lender is purely motivated by profits. In fact, it shares the objectives, risk attitudes, and intertemporal preferences of the competitive creditors who lend in debt markets. We make this assumption not to represent any specific bilateral lender accurately, but rather to isolate the pure effect of market structure. However, the feedback from debt levels to the bilateral surplus, which gives rise to the cross-elasticity and the relational overborrowing effect, will be present unless the large lender has a strong motivation to avoid default on the government's marketable debt. In this sense, the core results are robust to alternative descriptions of large bilateral creditors, including those with mixed policy, commercial, or financial stability objectives.

A key quantitative finding is that even relatively small bilateral loans can significantly affect sovereign debt outcomes when marketable debt has long maturity. The reason is that debt service on long-term debt is distributed over time, so the amount due in any given period is relatively small. By contrast, bilateral loans—modeled as short-term—must be repaid (or rolled over) every period. As a result, fluctuations in the bilateral loan interest rate have an impact on the current-period budget constraint that is comparable in magnitude to changes in the interest rate on a much larger stock of marketable debt.

The model delivers a simple heuristic for evaluating whether a particular form of bilateral lending is likely to be beneficial for the borrowing country. The key determinant is how the interest rate on bilateral loans responds to the level of the government's market debt: if this cross-elasticity is such that bilateral loan terms improve after larger debt issuances, the relational overborrowing channel is likely to be active, as the large lender's willingness to lend cheaply amplifies the government's incentive to accumulate market debt. By contrast, if the terms of bilateral lending become more favorable when debt levels are low or falling, it is more likely to support debt sustainability and improve welfare. These effects operate alongside more familiar channels related to avoiding improvements in the value of default, as recognized in policies like the IMF's Financing Assurances and Sovereign Arrears framework (IMF, 2024a), which aim to limit (bilateral) financing before at least a "credible process for debt restructuring is underway."

Discussion of the Literature We contribute to a nascent literature on the interaction of different types of sovereign debt.[3] Hatchondo, Martinez, and Önder (2017) find that introducing a limited amount of non-defaultable debt improves the government's welfare but only for a short period of time, while Cordella and Powell (2021) show how the preferred creditor status of international financial institutions can arise endogenously. Dellas and Niepelt (2016) focus on the coexistence of official and private creditors and show how an interior solution featuring both types of lending can arise as a choice of the borrowing government. Boz (2011) and Fink and Scholl (2016) study the interaction of market debt with a type of senior debt coming with conditionality. Kirsch and Rühmkorf (2017) and Roch and Uhlig (2018) investigate the role of official lending or bailout agencies in eliminating equilibrium multiplicity, as do Corsetti, Guimarães, and Roubini (2006). The model we consider does not feature this type of equilibrium multiplicity, which would open the door to welfare gains of bilateral loans if they could help rule out unfavorable equilibria. Importantly, we also abstract from conditionality and focus on the impact of market design itself. Also related are Kovrijnykh and Szentes (2007) and Faria-e-Castro et al. (2024) who describe situations featuring strategic behavior of creditors in different contexts, and Dovis and Kirpalani (2023) who propose a design for a lender of last resort that improves the worst equilibrium.

More recently, Arellano and Barreto (2025) combine market and official debt in a model of partial default (Arellano et al., 2023) and find that competitive pricing and longer maturites, like those associated with bilateral lending from the Paris Club, endogenously make official debts less risky despite more favorable restructuring terms. Liu, Liu, and Yue (2025) show how the best subgame-perfect equilibrium of a sovereign debt model with market, bilateral, and multilateral creditors decentralizes the constrained-efficient allocation with imperfect information and moral hazard. We view these papers as emphasizing how certain institutions could potentially improve welfare in a world with both market and official debts, while our results exemplify the perils of bilateral sovereign debt. Also on the risks side, Kondo, Sosa-Padilla, and Swaziek (2025) study a sovereign debt model augmented by exogenous capital flows from a large lender (e.g. China) and focus on the risk that it demands repayment unexpectedly.

Empirically, Perks et al. (2021), and Bahaj and Reis (2021, 2023) document the network of Central Bank swap lines. Cesa-Bianchi et al. (2022) study swap lines among advanced economies,

[3] Excellent surveys of the broader literature on sustainable public debt and sovereign default can be found in handbook chapters by Aguiar and Amador (2014), Aguiar, Chatterjee, Cole, and Stangebye (2016), D'Erasmo, Mendoza, and Zhang (2016), and Martinez, Roch, Roldán, and Zettelmeyer (2023).

where they can serve a different purpose.

Layout The rest of the paper is structured as follows. Section 2 presents some motivating evidence on the increasing role of official external debt in emerging markets. Section 3 presents two minimal models, highlighting the emergence of the cross-elasticity and how it can increase incentives for debt dilution. Then, Section 4 describes the main model in which both types of debt coexist, while Section 5 analyzes its equilibrium, and Section 6 describes the extension with fixed rules for the terms of bilateral loans. Finally, Section 7 concludes.

2. Motivating Evidence

Figure 1 summarizes the evolution of official external debt across emerging-market economies, using the World Bank's International Debt Statistics (IDS). The data include all public and publicly guaranteed (PPG) external sovereign debt, measured in constant 2023 USD and aggregated across debtor countries. We distinguish between three broad official creditor categories: multilateral lenders (including the IMF and World Bank), Paris Club bilateral lenders,[4] and other bilateral creditors—primarily non-Paris Club official lenders such as China.

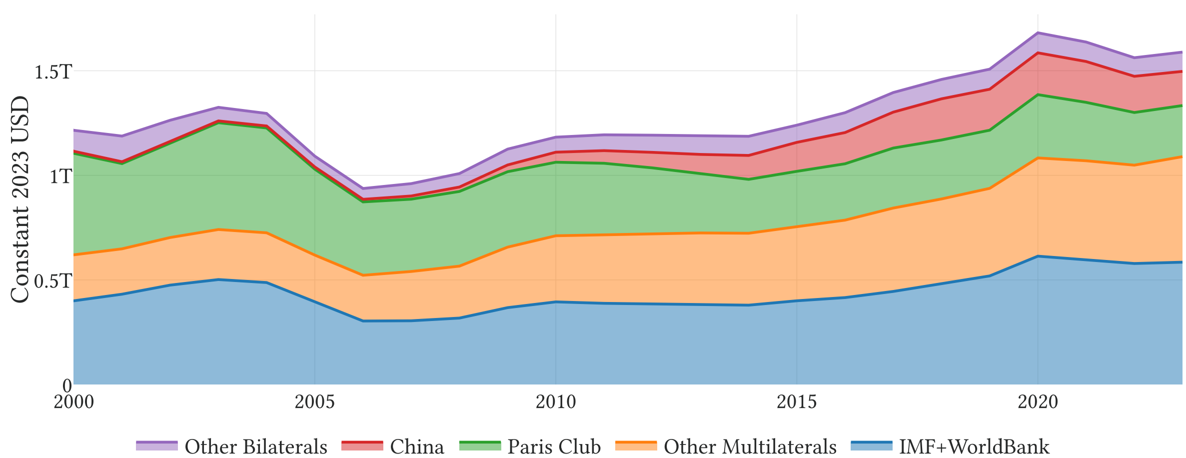

Figure 1: Total Official Debt

Source: International Debt Statistics, The World Bank

The data show a clear shift in the composition of official external debt over the past two

[4] As of 2025, the members of the Paris Club are Australia, Austria, Belgium, Brazil, Canada, Denmark, Finland, France, Germany, Ireland, Israel, Italy, Japan, Netherlands, Norway, Russia, South Korea, Spain, Sweden, Switzerland, the United Kingdom, and the United States.

decades. While the IMF and World Bank, along with Paris Club members, accounted for the bulk of official lending in the early 2000s, bilateral creditors outside the Paris Club have grown rapidly in both absolute and relative terms. By 2023, bilateral official lenders collectively account for roughly one-third of the global stock of official external debt to emerging markets, up from less than one-sixth two decades earlier. Among these, China has emerged as the single largest bilateral lender.

While the rise of new bilateral creditors is evident from Figure 1, it only reflects liabilities included within the official perimeters of PPG debt and hence reported to the World Bank. Arrangements with Central Banks (notably, swap lines) are often not included in such perimeters and therefore not captured in the figure.[5] This limitation in the data illustrates the opacity of some of these deals (other aspects, like interest rates and maturities, are also typically confidential), which underpins concerns about the welfare implications of these new lenders (Horn et al., 2021a, 2023). Moreover, the scarce data on certain lending channels underscores the need for analytical tools that go beyond reported debt stocks to capture strategic interactions in sovereign borrowing decisions. The rest of our paper addresses these concerns through a theoretical and quantitative evaluation of a model that incorporates a large bilateral lender operating alongside competitive market creditors.

3. Two Minimal Models

We begin our analysis by considering two stylized environments. These are designed to clarify the core mechanisms in the full model, described in Section 4. The first stylized model describes how the cross-elasticity emerges from the government's attempts to improve its bargaining position. The second stylized model takes the cross-elasticity as given and characterizes how it magnifies the incentives for debt dilution, potentially leading to welfare losses.

3.1 A Two-Period Model: How the Cross-Elasticity Emerges from Bargaining

A key feature of the quantitative model we introduce in Section 4 is a cross-elasticity of bilateral loan terms to market debt issuance. That full model features a within-period timing in which the government first chooses its market debt issuance and only afterward negotiates with the senior lender. As a result, the government can use market debt issuance to influence its cash on hand and, hence, its threat point at the bargaining stage.[6]

[5] Bahaj and Reis (2021, 2023) compile data on the network of Central Bank swap lines and, in many cases, are only able to see the total committed size of the swap lines, not drawn or borrowed amounts.

[6] Section 5.3 discusses how this type of threat-point manipulation can be captured under alternative timing assumptions.

The simplified setup in this section abstracts from market debt itself and instead takes the government's cash on hand as a primitive. This allows us to isolate the bargaining mechanism and characterize how cash on hand (or, more generally, initial resources) affects the government's threat point and, through it, the amount and interest rate on the bilateral loan.

There are two dates, $t = 0, 1$. The government receives endowments $y_0 < y_1$ and seeks to maximize an expected-utility objective with a strictly concave per-period utility function u and discount factor $\beta \in (0, 1)$:

$$V = u(c_0) + \beta u(c_1)\,.$$

The senior (and only) lender is risk-neutral and discounts time at $\beta_L \in (\beta, 1)$. All action occurs at the bargaining stage of period 0, when the government negotiates with the senior lender a transfer of x in exchange for a repayment of $m' = x(1 + r)$ due in period 1. The government is able to commit to this repayment. With a Nash bargaining weight of θ for the lender, the bargaining problem is

$$\max_{x,r} \mathscr{L}(x, r)^{\theta} \times \mathscr{B}(x, r; y_0, y_1)^{1-\theta} \tag{1}$$

where

$$\mathscr{L}(x, r) = -x + \beta_L x(1 + r) = x(\beta_L(1 + r) - 1)$$
$$\mathscr{B}(x, r; y_0, y_1) = \underbrace{u(y_0 + x) + \beta u(y_1 - x(1 + r))}_{\text{agreement}} - \Big(\underbrace{u(y_0) + \beta u(y_1)}_{\text{threat point}}\Big)$$

The focus of this section is to study the comparative statics of borrowing quantities x and terms r to the government's cash on hand y_0. As mentioned above, in the quantitative model, the government is able to affect its cash on hand by choosing its issuance of market debt. The strength of the cross-elasticity measures the incentive for the government to manipulate its threat point vis-à-vis the senior lender.

The solution to the bargaining problem (1) is characterized by two first-order conditions: an Euler equation,

$$u'(c_0) = \frac{\beta}{\beta_L} u'(c_1)\,,$$

and a surplus-sharing rule,

$$u'(c_0) = \frac{\theta}{1 - \theta} \frac{\mathscr{B}(x, r; y_0, y_1)}{\mathscr{L}(x, r)}\,,$$

where $c_0 = y_0 + x$ and $c_1 = y_1 - x(1 + r)$ are the on-path consumption levels.

The Euler equation means that the size of the transfer x should satisfy the borrower front-loading motive when using the lender's discount rate β_L^{-1} as the interest rate. In other words, the transfer is chosen to maximize total surplus. The surplus-sharing rule determines the terms at which the transfer is provided, setting the interest rate r to allocate surplus across lender and borrower.

Lemma 1. *Suppose the transfer size x is fixed at some $x^\star > 0$. Then, higher cash on hand y_0 leads to better terms when bargaining with the senior lender: $\frac{\partial r}{\partial y_0} < 0$.*

Proof of Lemma 1 In this case the first-order condition is

$$\theta\beta_L\mathscr{B}(x^\star, r; y_0, y_1) - (1-\theta)\beta\mathscr{L}(x^\star, r)u'(c_1) = 0$$

Omitting the arguments of $\mathscr{B}$ and $\mathscr{L}$ for readability, defining $F(r, y_0) = \theta\beta_L\mathscr{B} - (1-\theta)\beta\mathscr{L}u'(c_1)$, and applying the implicit function theorem, we have

$$\frac{\partial r}{\partial y_0} = -\frac{\partial F/\partial y_0}{\partial F/\partial r} = -\frac{\theta\beta_L\frac{\partial\mathscr{B}}{\partial y_0}}{\theta\beta_L\frac{\partial\mathscr{B}}{\partial r} - (1-\theta)\beta\frac{\partial\mathscr{L}}{\partial r}u'(c_1) + (1-\theta)\beta\mathscr{L}u''(c_1)x}$$

For given (x, r), strict concavity of u means that higher cash on hand y_0 reduces the government's surplus, so we have $\frac{\partial\mathscr{B}}{\partial y_0} < 0$, and the numerator is negative. For the denominator, higher interest rate r reduces the government's surplus, so $\frac{\partial\mathscr{B}}{\partial r} < 0$, and increases the lender's surplus, so $\frac{\partial\mathscr{L}}{\partial r} > 0$. Finally, the last term in the denominator is also negative, because of the concavity of u. In fact, the entire denominator being negative is the second-order condition for the maximization problem. We are then left with a negative numerator as well as a negative denominator, with a minus sign in front, which yields a negative sign for the derivative $\frac{\partial r}{\partial y_0}$. □

Lemma 2. *Suppose the interest rate r is fixed at some $r^\star > 0$. Then, higher cash on hand y_0 leads to a lower reliance on the senior lender: $\frac{\partial x}{\partial y_0} < 0$.*

Proof of Lemma 2 In this case the first-order condition is

$$\theta\beta_L\mathscr{B}(x, r^\star; y_0, y_1) - (1-\theta)\beta\mathscr{L}(x, r^\star)u'(c_1) = 0$$

Defining $F(x, y_0) = \theta\beta_L\mathscr{B} - (1-\theta)\beta\mathscr{L}u'(c_1)$, and applying the implicit function theorem once more, we have

$$\frac{\partial x}{\partial y_0} = -\frac{\partial F/\partial y_0}{\partial F/\partial x} = -\frac{\theta\beta_L\frac{\partial\mathscr{B}}{\partial y_0}}{\theta\beta_L\frac{\partial\mathscr{B}}{\partial x} - (1-\theta)\beta\frac{\partial\mathscr{L}}{\partial x}u'(c_1) + (1-\theta)\beta\mathscr{L}u''(c_1)(1+r)}$$

As in the previous case, we have $\frac{\partial\mathscr{B}}{\partial y_0} < 0$, and the denominator being negative is the second-order condition for the maximization problem, stating that the marginal benefit to the government of increasing the transfer is becoming lower than the marginal benefit to the lender as x increases above its optimal value. Once more, the relevant expression is the ratio of two negative numbers, with a minus sign in front. $\square$

When both x and r are free, the same force can be split across the two margins. Higher cash on hand lowers the surplus generated by any given agreement, so the Nash solution restores surplus sharing through a lower transfer, a lower interest rate, or both.[7] In the quantitative model, the relevant equilibrium responses move in this direction, as we show numerically in Section 5.

Discussion In the quantitative model, this cash-on-hand comparative static maps into the government's market debt choice. The government first chooses market debt issuance and then enters the bilateral negotiation. On the upward-sloping part of the debt Laffer curve, a higher choice of b' raises current resources and therefore strengthens the government's bargaining position vis-à-vis the large lender.[8] This is the origin of the cross-elasticity of bilateral loan terms to market debt issuance.

3.2 A Three-Period Model: How the Cross-Elasticity Worsens Debt Dilution

We turn to analyzing the effect of the cross-elasticity on debt dilution. Debt dilution is a well-known feature of models with long-term debt (Hatchondo, Martinez, and Sosa-Padilla, 2016), in which new issuances affect the price of marginal and inframarginal debt issued currently, but also of outstanding debt issued in the past. Because of this, with commitment the government would want to affect its future issuance strategy to boost current prices, leading to time inconsistency. In this section we show that the cross-elasticity magnifies the losses from this time-inconsistency problem: borrowing capacity with the large lender raises welfare for a government that can commit, but reduces it for a government that decides sequentially.

There are three dates, $t = 0, 1, 2$. In the final period, the government receives a stochastic endowment $y(z)$, where $z \sim N(0, \sigma_z^2)$, but wishes to consume in all periods to maximize an expected-utility objective

$$V = \mathbb{E}\left[\sum_{t=0}^{2} u(c_t)\right] \tag{2}$$

with an increasing and concave utility function u, satisfying Inada conditions.

[7]In principle, the effect of reducing one of them could be so large as to require an increase in the other variable. The conditions required to rule out these pathological cases seem to be quite weak and are satisfied for most parametrizations of this problem.

[8]Initial market debt works differently: legacy obligations reduce current resources and generally weaken the government's bargaining position. The cross-elasticity emphasized here comes from the effect of new issuance on resources available before the bilateral negotiation.

To finance consumption in all three periods, the government issues long-term bonds b at $t = 0$ as well as short-term bonds d and a bilateral loan m at $t = 1$. All three are expressed as promises to repay one unit of the good at $t = 2$. While the loan m is non-defaultable, the government faces a choice of whether to repay the bonds b and d, subject to some default costs $h(z)$. In other words, consumption in period 2 is given by

$$c_2 = \begin{cases} y(z) - m - (b + d) & \text{if repay} \\ y(z) - h(z) - m & \text{if default} \end{cases}$$

In this finite-horizon example, since the utility function is increasing, default occurs exactly in those states for which the costs of default fall short of the value of the debt. In other words,

$$c_2 = y(z) - m - \min\{b + d, h(z)\} \tag{3}$$

In the middle period, the government issues short-term debt d and loans m. Thanks to competition among risk-neutral investors, the price of debt reflects the probability of default, so $q(b, d) = \mathbb{P}(b + d < h(z)) = q(b + d)$, with a slight abuse of notation. Notice that the price of short-term debt also depends on the outstanding amounts of long-term debt (in a perfectly-substitutable way, as default affects both types of bonds equally).

In order to focus on how the cross-elasticity worsens the debt dilution problem, we assume an exogenous rule for the pricing of bilateral loans which takes the existence of the cross-elasticity as given: more legacy debt b may induce a lower price (higher interest rate) but more recently-issued debt d induces higher prices (the cross-elasticity), exactly as suggested by the bargaining model of Section 3.1. To keep things simple, we assume a linear pricing rule:

$$\phi(b, d, m) = \alpha_0 + \alpha_b b + \alpha_d d + \alpha_m m$$

where $\alpha_d > 0$ to represent in reduced form the negative response of the bilateral interest rate to cash on hand in Lemma 1. We also impose a fixed debt limit on the bilateral loan, $\bar{m}$. Consumption in the middle period is

$$c_1 = dq(b + d) + m\phi(b, d, m) \tag{4}$$

Finally, in the initial period, the government issues long-term debt b at price q, but investors hold beliefs d^e about the issuance of short-term bonds d, which will happen later. Initial consumption is

$$c_0 = bq(b + d^e) \tag{5}$$

for the same price function q as above (notice that, since the bilateral loan m does not affect the default set, there is no need to anticipate m to decide q).

We consider different cases for the government's ability to influence beliefs d^e.

Commitment When the government can commit, it chooses $\{b, m, d^e, d\}$ to maximize (2) subject to (3), (4), (5), $m \leq \bar{m}$ and $d = d^e$.

The first-order conditions for m (using λ to denote the multiplier on the debt limit constraint) and b are

$$u'(c_1)\left(\phi + m\frac{\partial \phi}{\partial m}\right) = \mathbb{E}\left[u'(c_2)\right] + \lambda \tag{6}$$

$$u'(c_0)\Big(bq'(b+d) + q(b+d)\Big) + u'(c_1)\left(dq'(b+d) + m\frac{\partial \phi}{\partial b}\right) = q(b+d)\mathbb{E}\left[u'(c_2^R)\right]$$

and the first-order condition for d is

$$u'(c_0)bq'(b+d) + u'(c_1)\left(dq'(b+d) + q(b+d) + m\frac{\partial \phi}{\partial d}\right) = q(b+d)\mathbb{E}\left[u'(c_2^R)\right] \tag{7}$$

where c_2^R denotes period-2 consumption conditional on repayment. Because of equal seniority of short- and long-term debt, the contribution of period 2 to the first-order conditions for b and d is the same. The loan, by contrast, is repaid in all states: its marginal cost in (6) is the unconditional expectation $\mathbb{E}\left[u'(c_2)\right]$, while the bonds' marginal cost is only incurred in repayment states.

Discretion When deciding sequentially, the government takes as given a value d^e and chooses $\{b, m, d\}$ to maximize (2) subject to (3), (4), (5) and $m \leq \bar{m}$. Equilibrium then requires investors to be correct in their assessment, so that the government does choose $d = d^e$. As a result, the first-order conditions for both b and m remain unchanged, but the first-order condition for d is now 'missing' the term reflecting the manipulation of d^e by the committed government:

$$u'(c_1)\left(dq'(b+d) + q(b+d) + m\frac{\partial \phi}{\partial d}\right) = q(b+d)\mathbb{E}\left[u'(c_2^R)\right] \tag{8}$$

How the cross-elasticity worsens debt dilution Debt dilution manifests itself in how different the issuance choices are under commitment and under discretion. We therefore compare three solutions, which Figure 2 reports as functions of the maximum amount $\bar{m}$ that can be drawn on the bilateral loan: the commitment solution (b_C, d_C); the equilibrium with discretion (b_D, d_D); and an intermediate object (b_{DC}, d_{DC}), defined as the choices of a sequential government that reoptimizes while bond prices still reflect commitment beliefs, $d^e = d_C$. The intermediate solution isolates the pure *temptation* to deviate from the commitment plan; the comparison between d_{DC} and d_D then isolates the amplification that comes from the equilibrium adjustment of prices. The result of this section is that the gap between d_C and d_D grows with $\bar{m}$: more bilateral borrowing capacity can worsen dilution, hurting welfare.

A useful benchmark is the commitment solution with $\bar{m} = 0$. With no discounting and the same default technology applying to both bonds, the committed government treats the two issuance periods symmetrically: the cross-price terms $u'(c_0)\,bq'$ and $u'(c_1)\,dq'$ appear in both (6)

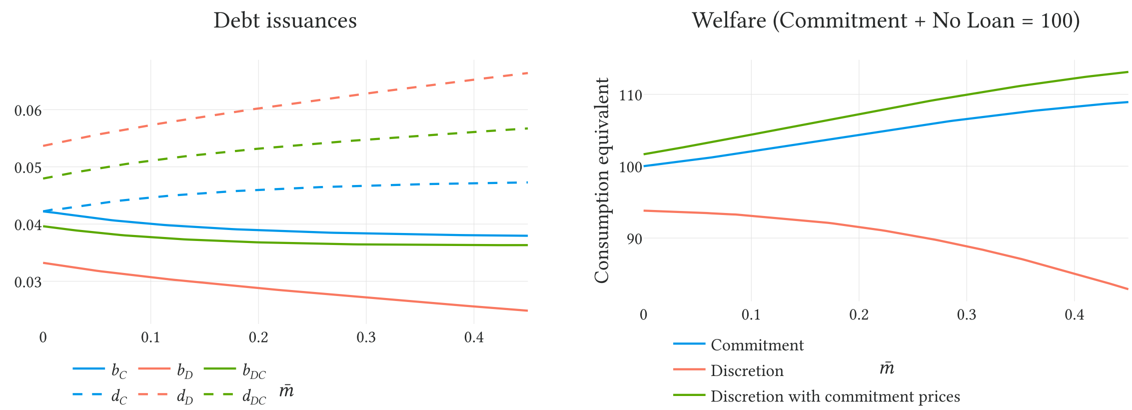

FIGURE 2: DEBT DILUTION AND THE CROSS-ELASTICITY

Note: The figure assumes $(\alpha_0, \alpha_b, \alpha_d, \alpha_m) = (-0.025, -1.5, 2.5, 0)$, a CRRA utility function with risk aversion $\gamma = 2$, income volatility $\sigma_z = 10\%$, and output costs of default $h(z) = 0.1 \cdot y(z)$

and (7), so subtracting one condition from the other leaves

$$u'\Big(\underbrace{q(b+d)\,b}_{=c_0}\Big)\,q(b+d) = u'\Big(\underbrace{q(b+d)\,d}_{=c_1}\Big)\,q(b+d) \qquad \Longrightarrow \qquad c_0 = c_1 \;\text{ and }\; b = d\,.$$

Against this benchmark, the committed and the sequential government differ in exactly one term of the first-order condition for d: comparing (7) with (8), the sequential government does not perceive the *dilution wedge*

$$\Delta \equiv -u'(c_0)\,b\,q'(b+d) > 0\,, \tag{9}$$

the marginal damage that anticipated period-1 issuance inflicts on the period-0 price of long-term debt. The committed government internalizes Δ and stops issuing d while the flow benefit of issuance still exceeds its period-2 cost by exactly Δ. The sequential government lacks the wedge Δ in its first-order condition and compensates by increasing d to restore equality between marginal benefits and costs. It therefore issues more, and this comes at the expense of b: the higher default probability worsens debt prices and induces the government to curb long-term issuance. At $\bar{m} = 0$, the gap $d_D - d_C$ measures the extent of debt dilution coming only from the presence of long-term debt.

The bilateral loan magnifies these effects. Suppose the borrowing limit binds, so that $m = \bar{m}$.[9] More m raises period-1 resources directly through the proceeds $m \cdot \phi$ as well as indirectly because, with $\alpha_d > 0$, each unit of d now also improves the price ϕ and adds $m \cdot \alpha_d$. These extra resources at $t = 1$ lower the marginal utility $u'(c_1)$. The wedge Δ, in contrast, is predetermined and thus

[9] If the limit does not bind, the cross-elasticity still raises loan usage under discretion: the larger d pushes ϕ up and, through (6), induces $m_D > m_C$.

unaffected by m. Hence, the compensating effect of d on the first-order condition (8) is weaker, because the wedge is unchanged but the effect of d, which is scaled by $u'(c_1)$, is lower. As a consequence, the sequential government relies on quantities and chooses a larger d as m grows. The temptation to dilute, measured by $d_{DC} - d_C$ (ignoring for now the equilibrium response of expectations), is therefore increasing in m, as the left panel of Figure 2 confirms.

Equilibrium expectations amplify the deviation further. When investors anticipate the higher issuance, d^e rises and the period-0 price of long-term bonds falls; the government responds by reducing b, offsetting this effect only partially. Lower prices pull time-1 issuance in two directions: each bond raises less revenue, which discourages issuance (a substitution effect), but the implied fall in revenues raises the marginal utility of period-1 consumption, which encourages issuing more (an income effect). With CRRA utility and relative risk aversion above one, the income effect dominates and the equilibrium response adds to the deviation: $d_D > d_{DC}$, with this gap also widening in $\bar{m}$.[10]

The right panel of Figure 2 summarizes the welfare implications. Under commitment, the bilateral loan is unambiguously valuable, as a direct consequence of free disposal. The value of reoptimizing while prices still reflect commitment beliefs is even higher, reflecting the time-inconsistent nature of the problem. Once investors anticipate this deviation, which is increasing in m as described above, equilibrium welfare under discretion falls as the debt-dilution problem worsens. The quantitative model of Section 4 embeds this mechanism in an infinite-horizon setting in which the cross-elasticity emerges endogenously from bargaining.

4. Model with bilateral loans and defaultable market debt

In this section we present the full version of the model, in which the borrowing government has access to the large lender as well as to a competitive fringe of bondholders. Default on the debt b held by competitive lenders is possible, subject to standard output costs of default. However, just as before, bilateral loans m cannot be defaulted.

The timeline of events within a period is illustrated in Figure 3.

At the start of t, the government owes m to the large lender, b to the market, and observes the exogenous state z. Additionally, the economy can be in default ($\zeta = D$) or in repayment ($\zeta = R$). Let $v(b, m, z)$ and $h(b, m, z)$ represent the government's and the large lender's value functions at the beginning of a period, when the government is in good standing with the market.

[10] The coefficient α_b matters for the commitment benchmark itself. In the quantitative model, legacy market debt weakens the government at the bargaining stage, which maps into $\alpha_b < 0$ here. In that case a larger m tilts even the committed government away from long-term and toward short-term issuance, consistent with the behavior of b_C and d_C in Figure 2.

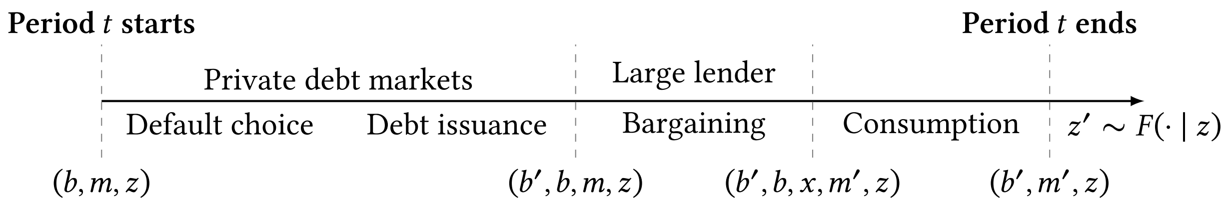

FIGURE 3: TIMELINE OF EVENTS WHILE NOT IN DEFAULT

The government accesses the market at the beginning (morning) of each period t and chooses whether to default or repay. If it chooses to repay, the government can issue new debt. Later, in the afternoon of period t, the government contacts the large lender and bargains over a transfer as before.

Private markets In the morning of t, first, the government decides default for the current period if it is in good standing with the market.

$$v(b, m, z) = \max \{v_R(b, m, z) + \epsilon_R, v_D(m, z) + \epsilon_D\} , \tag{10}$$

where the ϵ's follow a Type 1 Extreme Value distribution, yielding closed forms for $v(b, m, z)$ and the (beginning-of-period) default probability $P(b, m, z)$

$$v(b, m, z) = \chi \log \big(\exp(v_D(m, z)/\chi) + \exp(v_R(b, m, z)/\chi)\big)$$
$$P(b, m, z) = \frac{\exp(v_D(m, z)/\chi)}{\exp(v_D(m, z)/\chi) + \exp(v_R(b, m, z)/\chi)}$$

We assume that defaulting on the market debt fully destroys all bonds outstanding and triggers exclusion from international capital markets. Access to markets is then recovered stochastically with constant probability ψ. In addition, the maximum amount that can be borrowed from the large lender m' while in default on market debt is constrained by $m' \le \Gamma(m)$. We begin our analysis without restrictions by setting $\Gamma(m) = +\infty$ but describe below different possibilities for the value of Γ.

If default is not chosen in the current period, the government issues new debt b' to the fringe of lenders understanding the value of entering negotiations with the senior lender after having issued this debt level b',

$$v_R(b, m, z) = \max_{b'} w_R(b', b, m, z) , \tag{11}$$

where debt is a perpetuity of geometrically-decaying coupons (Leland, 1998; Hatchondo and Martinez, 2009; Arellano and Ramanarayanan, 2012). A unit of debt issued in period t promises to repay $\kappa(1-\delta)^{s-1}$ units of the tradable good in period $t + s$, effectively making a unit issued at $t - 1$ a perfect substitute for $(1 - \delta)$ units issued at t. In this sense, δ controls the maturity of debt while κ controls the average size of the coupon payments.

The price faced by the borrower government reflects its lenders' expectations of repayment, discounted with a risk-neutral kernel

$$q(b', b, m, z) = \frac{1}{1+r^*}\mathbb{E}\left[(1 - \mathbb{1}_D(b', m', z'))\left(\kappa + (1-\delta)q(b'', b', m', z')\right) \mid z\right] \tag{12}$$

where r^* is the international risk-free rate,[11] $\mathbb{1}_D(b, m, z)$ denotes the government's default policy as perceived by the competitive lenders, $m' = m'_R(b', b, m, z)$ is the expected result of negotiations with the large lender, to happen later in the period, and $b'' = g_b(b', m'_R(b', b, m, z), z')$ is the expected debt issuance in the following period.

Bilateral loan In the afternoon of t, the government meets with the large lender to negotiate the bilateral loan. As before, the outcome of their negotiation is a transfer x and new loan size m' which solve the following Nash bargaining problems, depending on default status vis-à-vis the market

$$\begin{gathered}\max_{m',x} \mathscr{L}_R(b', x, m, m', z)^{\theta}\ \mathscr{B}_R(b', b, x, m, m', z)^{1-\theta} \\ \text{or} \\ \max_{m',x} \mathscr{L}_D(x, m, m', z)^{\theta}\ \mathscr{B}_D(x, m, m', z)^{1-\theta}\end{gathered} \tag{13}$$

As before, the large lender's surplus is

$$\begin{aligned}\mathscr{L}_R(b', x, m, m', z) &= -x - m + \beta_L\mathbb{E}\left[h(b', m', z') - h(b', 0, z') \mid z\right] \\ \mathscr{L}_D(x, m, m', z) &= -x - m + \beta_L\mathbb{E}\left[\psi\left(h(0, m', z') - h(0, 0, z')\right) + (1-\psi)\left(h_D(m', z') - h_D(0, z')\right) \mid z\right],\end{aligned}$$

while the borrower's surplus now also reflects outcomes in debt markets

$$\begin{aligned}\mathscr{B}_R(b', b, x, m, m', z) &= u(y(z) + B(b', b, m, z) + x) - u(y(z) + B(b', b, m, z) - m) + \\ &\quad + \beta\mathbb{E}\left[v(b', m', z') - v(b', 0, z') \mid z\right] \\ \mathscr{B}_D(x, m, m', z) &= u(y_D(z) + x) - u(y_D(z) - m) + \\ &\quad + \beta\mathbb{E}\left[\psi\left(v(0, m', z') - v(0, 0, z')\right) + (1-\psi)\left(v_D(m', z') - v_D(0, z')\right) \mid z\right]\end{aligned}$$

where the function $y_D(z) = y(z) - \xi(z)$ is output in default and $B(b', b, m, z)$ summarizes net transfers from the competitive lenders received in the morning. Our assumptions on debt maturity yield $B(b', b, m, z) \equiv q(b', b, m, z)(b' - (1-\delta)b) - \kappa b$. In default, opportunities to reaccess markets arrive with probability ψ. The bargaining problems yield outcomes for the bilateral loan $\left\{x_R(b', b, m, z), m'_R(b', b, m, z)\right\}$ and $\left\{x_D(m, z), m'_D(m, z)\right\}$ in repayment and in default, respectively.

This bargaining problem, which is cast in terms of a transfer $x \geq -m$ in exchange for a new balance m', implies an interest rate r which satisfies the identity

$$(x + m)(1 + r) = m', \tag{14}$$

[11] We set $\kappa = r^* + \delta$, so that the price of debt equals 1 when the government repays with certainty.

which maps into the stylized setup of Section 3.1.

The most important way in which the presence of debt markets affects the bargaining stage is through the net flows from debt repayment and new issuance, $B(b', b, m, z)$. These flows modulate the government's threat point: after a successful issuance which raises large revenues, the government is in a strong position to negotiate as repaying m is less costly.

Consumption After the negotiation is done and transfers settled, consumption takes place.

$$\begin{aligned} c_R(b', b, m, z) &= y(z) + B(b', b, m, z) + x_R(b', b, m, z) \\ c_D(m, z) &= y(z) - \xi(z) + x_D(m, z) \end{aligned} \tag{15}$$

Given transfers $x_R(b', b, m, z)$ and $x_D(m, z)$, we can now return to the beginning of the period and determine the values of repaying and defaulting

$$\begin{aligned} w_R(b', b, m, z) &= u(c_R(b', b, m, z)) + \beta \mathbb{E}\left[v\left(b', m'_R(b', b, m, z), z'\right) \mid z\right] \quad \text{and} \\ v_D(m, z) &= u(c_D(m, z)) + \beta \mathbb{E}\left[\psi v\left(0, m'_D(m, z), z'\right) + (1-\psi) v_D\left(m'_D(m, z), z'\right) \mid z\right] , \end{aligned} \tag{16}$$

while for the large lender we have

$$\begin{aligned} h(b, m, z) &= P(b, m, z) h_D(m, z) + (1 - P(b, m, z)) h_R(b'(b, m, z), b, m, z) , \\ h_R(b', b, m, z) &= -x_R(b', b, m, z) + \beta_L \mathbb{E}\left[h(b', m'_R(b', b, m, z), z') \mid z\right] , \text{ and} \\ h_D(m, z) &= -x_D(m, z) + \beta_L \mathbb{E}\left[\psi h(0, m'_D(m, z), z') + (1-\psi) h_D(m'_D(m, z), z') \mid z\right] . \end{aligned} \tag{17}$$

5. Quantitative results

We parametrize our model at a quarterly frequency following standard strategies in the sovereign default literature. Most parameters are set externally, such as those governing the stochastic process for income, risk-free rates, risk aversion, the length of exclusion after default, and the duration of market debt (Chatterjee and Eyigungor, 2012).[12]

The critical parameter θ, which governs the lender's bargaining power, is set to 0.5 in our baseline but we study comparative statics around it later on. The remaining parameters (the sovereign's discount rate and the default costs) are calibrated internally to match the average debt-to-output ratio as well as the average and standard deviation of spreads. Our calibration is based on comparing data for Argentina between 1993q1 and 2001q4 to simulated data from the model. For the simulations, we use 35-period samples that end just before a default event, and condition on no default occurring within those 35 quarters or two years prior, as is standard in this

[12] As has also become standard for numerical performance, we include preference shocks for default as described above, but also for the debt issuance choice; both are set to a small variance to avoid impacting the equilibrium.

literature when assuming no recovery after default (Arellano, 2008; Chatterjee and Eyigungor, 2012).

Tables 1 and 2 summarize our parametrization and calibration.

TABLE 1: BASELINE PARAMETER VALUES

	Parameter	**Value**
Sovereign's discount factor	β	0.9607
Sovereign's risk aversion	γ	2
Preference shock scale: default	χ	0.025
Preference shock scale: issuance	χ_b	0.005
Lender's bargaining power	θ	0.5
Risk-free interest rate	r	0.01
Duration of debt	ρ	0.05
Income autocorrelation coefficient	ρ_z	0.9484
Standard deviation of y_t	σ_z	0.02
Reentry probability	ψ	0.0385
Default cost: linear	d_0	-0.2514
Default cost: quadratic	d_1	0.292

TABLE 2: TARGETED MOMENTS

	Data	**Only market**
Avg spread (bps)	815	740
Std spread (bps)	443	485
Debt to GDP (%)	17.4	18.1

Note: Simulation moments are based on 35-quarter samples before a default on market debt, conditional on no default occurring within those 35 quarters or two years prior, in a version of the model without bilateral loans. Data moments are computed for Argentina between 1993q1 and 2001q4.

Table 3 presents some statistics from simulating the model with and without bilateral loans, for different values of the lender's bargaining weight θ. Statistics correspond to the ergodic distribution of each model and the term 'loans' refers to bilateral loans m. Welfare gains are computed by comparing the economy with only market debt with each alternative model but setting $m = 0$, calculated averaging the ergodic distribution of the equilibrium in which only market debt is allowed, conditional on repayment, and reported in equivalent consumption terms. The left-most column corresponds to the equilibrium with only market debt upon which our calibration is based.[13]

Table 3: Business cycle statistics with and without bilateral loans

	Only market	**Unrestricted,** $\theta = 0.25$	**Unrestricted,** $\theta = 0.5$
Avg spread (bps)	630	1,005	1,398
Std spread (bps)	474	620	894
$\sigma(c)/\sigma(y)$ (%)	104	99.5	99
Debt to GDP (%)	16.4	17.7	17.6
Loan to GDP (%)	0	2.79	2.44
Loan spread (bps)	–	–	–
Corr. loan & spreads (%)	–	64.2	68.9
Default frequency (%)	5.03	7.67	9.55
Welfare gains (rep)	–	0.17%	-0.095%

Note: Statistics are based on simulations of the ergodic distribution of each model, conditional on repayment. The column marked 'Only market' refers to a version of the model without bilateral loans, while the columns marked 'Unrestricted' refer to versions in which bilateral loans can be used even during default episodes ($\Gamma(m) = +\infty$). Welfare gains are given as equivalent increases in consumption from the model with only market debt to the full model at $m = 0$, calculated averaging over the ergodic distribution of the model with only market debt.

The remaining columns show that even with a relatively large bargaining weight for the government, the availability of bilateral loans substantially increases the frequency of default. As a consequence, the government pays higher and more volatile spreads. In both cases, the government is able to place more market debt. This combined with the large lender's financing allows more frontloading of consumption, which the relatively impatient government values. When the large lender's bargaining power is low ($\theta = 0.25$), the increase in amounts borrowed is enough to overcome the worsening of the equilibrium in terms of default frequency and spreads.

Since the large lender keeps a share of the surplus generated by the loan, the borrower government is somewhat reluctant to use it. In a typical simulation path, conditional on no default the amount borrowed bilaterally is 2.4% of annual income with a standard deviation of 1.5%. Figure 4 shows that this changes significantly around default events: the loan size m shoots up before

[13] Statistics on Table 3 are slightly different than those in Table 2 which is based on the pre-default samples used for calibration.

the moment of default.

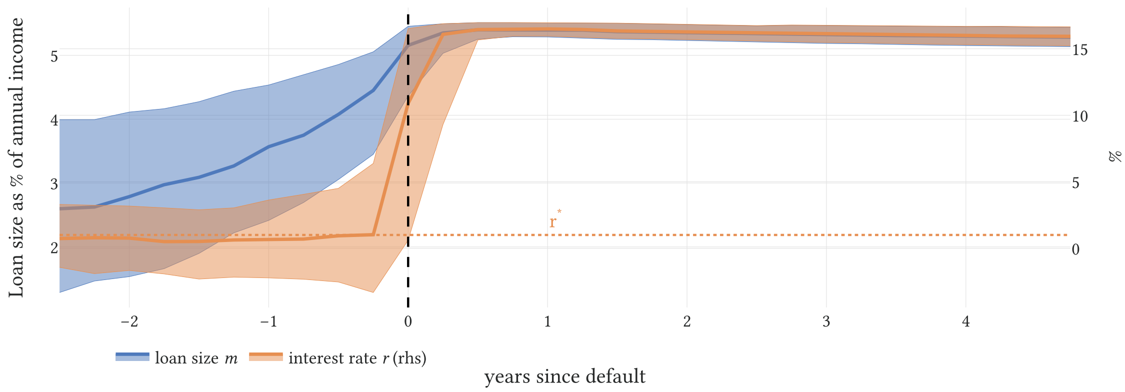

Figure 4: Loans around default events

Note: Event studies around default of market debt, with Unrestricted ($\Gamma(m) = +\infty$) bilateral loans and $\theta = 0.5$.

The large lender subsidizes this accumulation of bilateral debt, to then raise rates during the default spell. This behavior includes some risk-taking by the large lender: when the economy recovers market access, it immediately issues market debt to pay off the bilateral loan (see Figure 13 in the Appendix). The large lender is gambling on the exclusion period being long.[14]

While most of the use of the bilateral loan m occurs during default, Figure 4 shows that default episodes are preceded by bilateral issuance in an effort to avoid or postpone default (Figure 4 does not show the defaults that were avoided as a consequence of the large lender's presence).

Bilateral loans affect the economy in two ways: on the one hand, they provide extra financing when default risk makes borrowing in private markets costly. But they also provide funds in default, which raises the value of being excluded from private markets and consequently spreads. To disentangle these effects, we consider variants of the model in which the large creditor is less willing to lend when the economy is not in good standing with private markets. We capture this possibility via the constraint for the bargaining problem which requires that $m' \leq \Gamma(m)$ while the government is in default. Notice that the Unrestricted version discussed so far corresponds to the case of $\Gamma(m) = \infty$. We consider two variants. One in which bilateral loans are Unavailable during default, such that $\Gamma(m) = 0$. In this version, the government must repay any existing balances in its bilateral loan upon defaulting, which effectively constitutes an extra cost of default. Second, a Limited version in which $\Gamma(m) = m$, so that existing bilateral loans can be rolled over during default, but new net financing is off the table. In what follows we present results for all three

[14]This behavior also explains the negative average spread on the bilateral loan reported in Table 3. That statistic is a time average. The expected profits of the large lender correspond to an average of spreads weighted by borrowed amounts, and these two are positively correlated.

versions but focus on the Limited version.

Table 4 shows how these versions compare to the Unrestricted version and the equilibrium without bilateral loans, along their ergodic distributions.

Table 4: Business cycle statistics with bilateral loans

	Only market	**Unrestricted,** $\Gamma(m) = +\infty$	**Limited,** $\Gamma(m) = m$	**Unavailable,** $\Gamma(m) = 0$
Avg spread (bps)	630	1,398	1,209	650
Std spread (bps)	474	894	860	381
$\sigma(c)/\sigma(y)$ (%)	104	99	100	104
Debt to GDP (%)	16.4	17.6	17.6	18
Loan to GDP (%)	0	2.44	2.22	0.903
Loan spread (bps)	–	-298	-126	523
Corr. loan & spreads (%)	–	68.9	67.7	63.8
Default frequency (%)	5.03	9.55	9.04	5.28
Welfare gains (rep)	–	-0.095%	-0.038%	0.12%

Note: Statistics are based on simulations of the ergodic distribution of each model, conditional on repayment. 'Only market' refers to a version of the model without bilateral loans; all others are computed with $\theta = 0.5$. Default frequencies are computed on the ergodic distribution, as are welfare gains, which are given as equivalent increases in consumption calculated over the ergodic distribution of the model without bilateral loans, conditional on repayment.

The version in which bilateral loans are Limited during default removes some of the more mechanical incentives to default, as there is no possibility to obtain financing from the large lender to smooth or compensate the output costs of default. However, the decrease in default frequencies that obtains is rather small, with spreads still significantly higher than in the case with market financing only. When bilateral loans are fully Unavailable during default, they act as an extra cost of default, as any balance must be repaid in full immediately upon defaulting. In this version it can also be seen that the government becomes much more reluctant to borrow from the large lender. In this case, the default frequency is still slightly higher than in the case without bilateral loans, but the government is able to place a bit more debt (both market and bilateral), which pushes welfare up.

5.1 Default probabilities and debt prices

Figure 5 shows ex-post default regions for private debt, when the bilateral loan $m = 0$. Solid lines represent the case without bilateral loans, while dotted and dashed lines correspond to the versions with Unrestricted, Limited, and Unavailable loans. By and large, the presence of bilateral loans does not change debt capacity, with only marginal changes in the default sets. When loans are Unrestricted, the default region expands marginally, but not to an extent that would justify the increase in default frequency reported in Table 4. When loans are Limited, debt capacity actually increases marginally, while the default frequency is almost as large as in the Unrestricted case. Finally, when loans are Unavailable during default, the default region is virtually unchanged relative to the case without loans. These results point to a different culprit for the increase in default frequencies when bilateral loans are introduced.

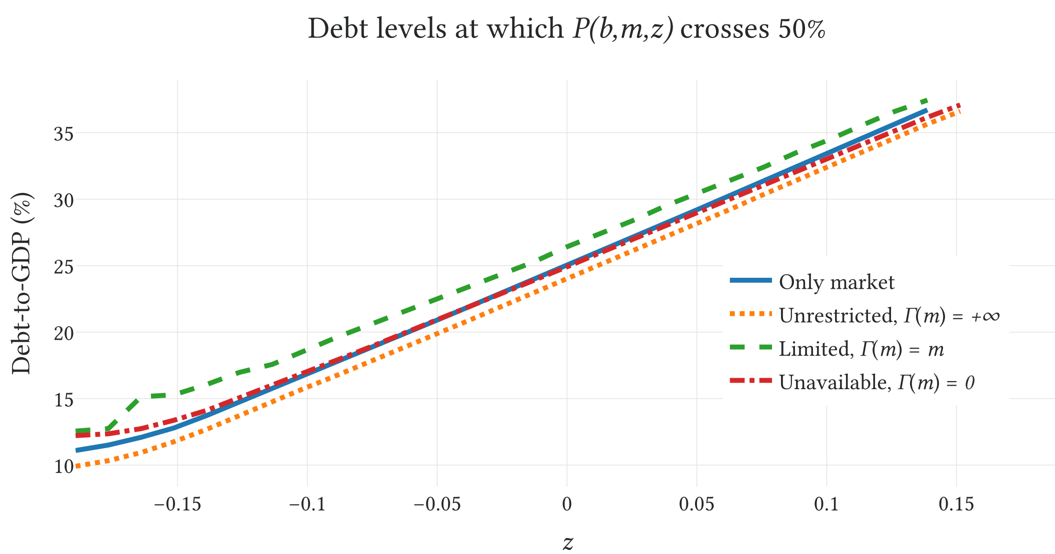

Figure 5: Default regions

Note: Default regions computed at $m = 0$, all models with bilateral loans parametrized with $\theta = 0.5$.

Figure 6 shows debt prices q as a function of the debt choice b' when the government owes nothing to either type of lender (i.e. $b = m = 0$). It highlights the negative impact of bilateral loans and the main consequence of the increase in default frequencies that we observe when the large lender is present. In the Unrestricted and Limited versions, market debt prices are depressed relative to the case without bilateral loans, especially when debt is low (in the Limited case, there is an intermediate debt level which prices are improved). Even though the one-period-ahead default probability does not increase substantially when bilateral loans are introduced, debt prices, which reflect a present value over a longer horizon, do fall, suggesting that debt dilution is increased. Finally, the case when loans are Unavailable in default is interesting, as spreads are *lower* at all debt levels, while the average spread level and the default frequency reported in Table

4 were higher.

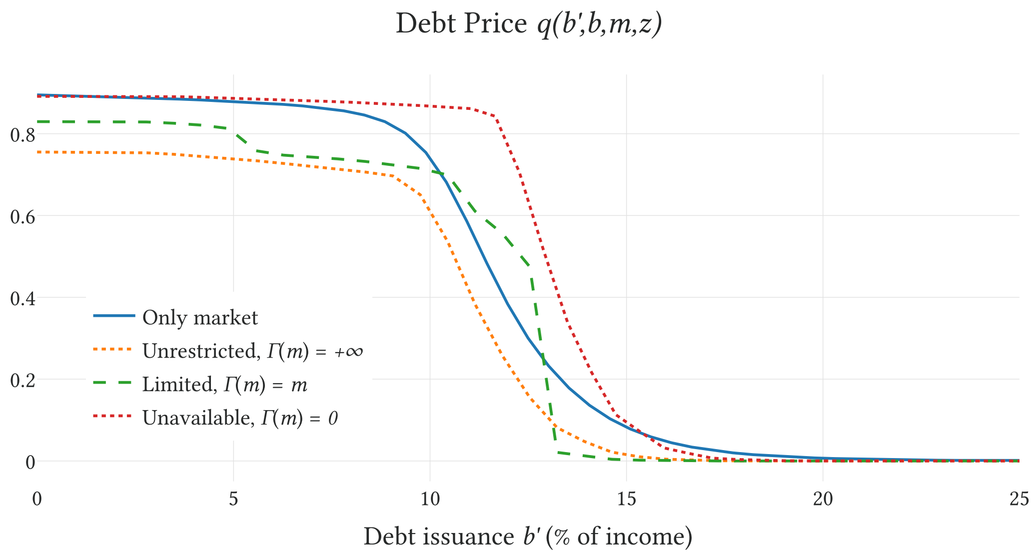

FIGURE 6: DEBT PRICES

Note: Debt prices computed at $b = m = 0$, all models with bilateral loans parametrized with $\theta = 0.5$.

5.2 Dynamics with bilateral loans

In the model with bilateral loans, the government issues debt in a riskier manner. Figure 7 presents the ergodic distribution of the debt-to-GDP ratio in simulations of the four models, conditional on repayment. When bilateral loans are available, in all variants, the economy spends more time in the region of the state space where default risk is large.

Figure 7 contains the key to the increase in default frequencies and spreads when the large lender is present. While debt capacity is not affected by the availability of bilateral loans, the government does spend much longer in the region of the state space where default risk is heightened. This also explains why debt prices are depressed (everywhere) even as the one-period-ahead default probability, which is also a state-by-state object, does not increase substantially upon the introduction of bilateral loans. We now investigate why the equilibrium with the large lender present is characterized by this more aggressive borrowing behavior by the government.

5.3 Relational overborrowing

The key to the government's market borrowing decisions in the morning lies in the afternoon. At the bargaining stage, the choice of market debt b' has already been made by the government.

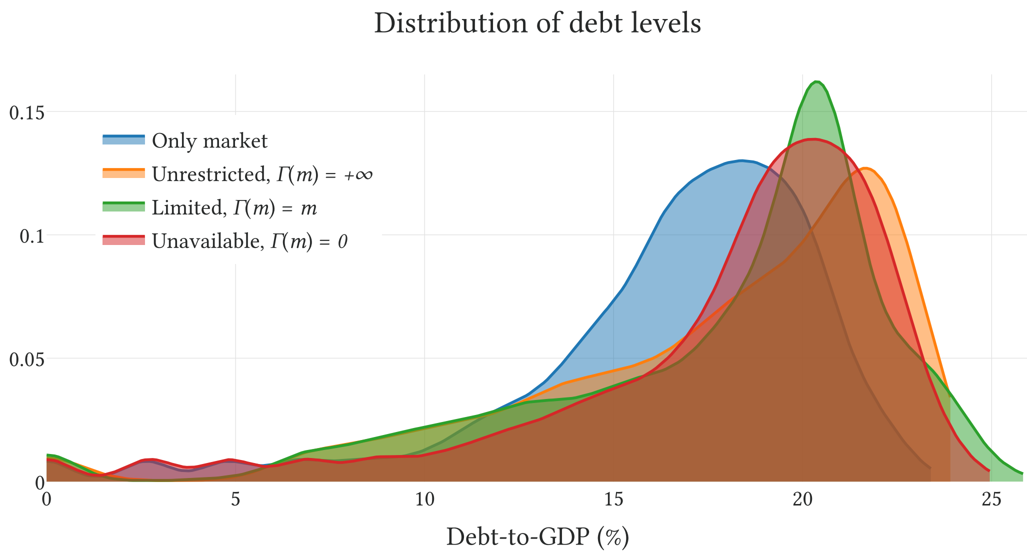

Figure 7: Distribution of debt levels

Note: All models with bilateral loans parametrized with $\theta = 0.5$.

As a result, if bilateral negotiations break down and the threat point needs to be exercised, the borrower is left with a consumption level of $y(z) + q(b', b, m, z)(b' - (1-\delta)b) - \kappa b - m$.

The term $B(b', b, m, z) = q(b', b, m, z)(b' - (1-\delta)b) - \kappa b$ captures net financing received from the market in the morning. It critically affects the government's threat point at the bargaining stage. In periods when the government is trying to reduce market debt b', net financing B is very low (or negative), and the government enters negotiations with the large lender in a 'weak' position as consumption at the threat point is low. This generates large gains (for the borrower) from any transfer x provided by the large lender. Thus, in order to split the surplus such transfers must come at a high interest rate. Conversely, when the government has received a large amount of net financing B, it expects a high level of consumption even if negotiations break down, which results in a strong position and good borrowing terms with the large lender. In the language of the stylized model of Section 3.1, net financing B increases y_0 and decreases y_1, strengthening the government's bargaining position and resulting in better terms from the large lender. The result is a negative cross-elasticity of the bilateral loan's interest rate to net financing B.

Figure 8 shows the interest rate that results from bilateral negotiations as a function of the choice of b',[15] when b is at an intermediate level (marked by the vertical dashed line) and $m = 0$. It confirms the discussion above: bilateral borrowing terms quickly worsen when the government delevers in debt markets.

As a result of this elasticity of loan terms to debt, new terms appear in the government's Euler

[15]At state (b, m, z), the choice of b' fully determines debt prices $q(b', b, m, z)$ and net financing $B(b', b, m, z)$.

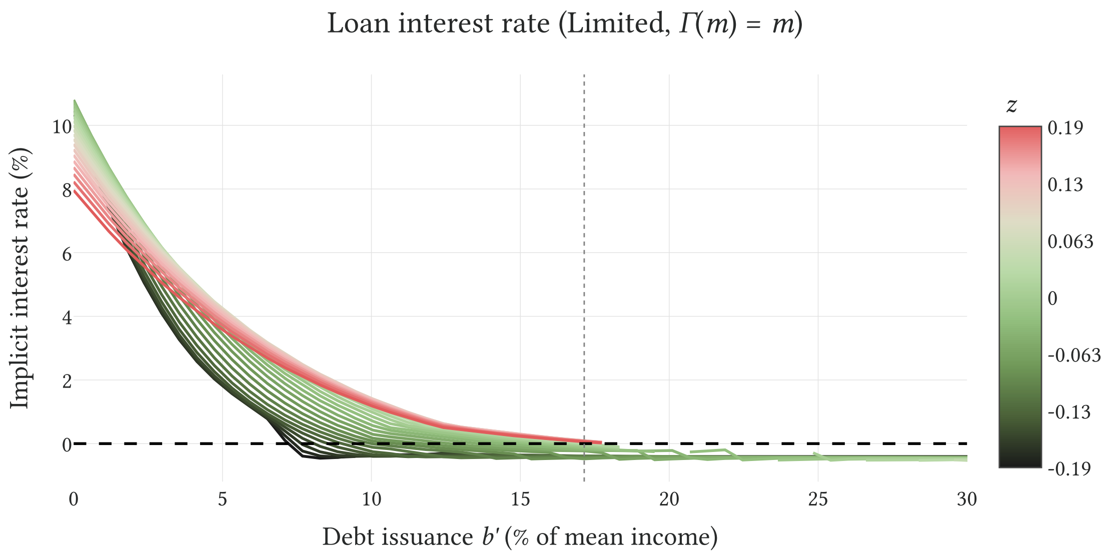

FIGURE 8: INTEREST RATE ON THE BILATERAL LOAN

Note: Interest rate on the bilateral loan computed at $m = 0$, b at 17% of mean income, for Limited bilateral loans ($\Gamma(m) = m$) and $\theta = 0.5$.

equation for market debt b':

$$u'(c)\left(q + \frac{\partial q}{\partial b'}i + \frac{\partial x}{\partial b'}\right) = \beta\mathbb{E}\left[u'(c')(1 - \mathbb{1}_D)\left(\kappa + (1-\delta)q' - i'\left(\frac{\partial q'}{\partial b} - \frac{\partial q'}{\partial m}\frac{\partial m'}{\partial b'}\right) - \frac{\partial x'}{\partial b} + \frac{\partial x'}{\partial m}\frac{\partial m'}{\partial b'}\right)\right] \tag{18}$$

where $i = b' - (1-\delta)b$ represents new market debt issuances, $\frac{\partial q'}{\partial b}$ refers to the derivative of the following period's debt price with respect to the beginning-of-period debt level at that point (in other words, b' from the perspective of the current period). Similarly, the term $\frac{\partial q'}{\partial m}\frac{\partial m'}{\partial b'}$ captures the derivative of the following period's debt price q' to the amount owed to the large lender at that moment (next period's m, which is today's m'), times the derivative of the current choice of m' to the current choice of b', and similarly for $\frac{\partial x'}{\partial b}$ and $\frac{\partial x'}{\partial m}$. Relative to a world with only market debt, there are three new effects arising from the presence of the large lender.

First, issuing debt affects transfers x received from the large lender in the afternoon of the current period. Second, issuing debt also affects the desired amount of loans m' taken from the large creditor. This flows into debt prices q' and transfers x' in the following period through the chain-rule terms. Finally, because debt prices and borrowing terms also depend on the initial level of indebtedness b, issuing new debt affects both the price at which market debt will be placed q' as well as transfers x' in the following period.

The left-hand side of (18) describes a countervailing force to the market discipline of spreads. In a model with only market debt, the benefit of issuing an extra unit of debt is $u'(c)\left(q + \frac{\partial q}{\partial b'}i\right)$: the government understands that issuing more debt may increase spreads (decrease the price q)

and this mitigates issuances. With the large lender, the benefit of issuing an extra bond includes additionally the term $\frac{\partial x}{\partial b'}$ as the extra issuance also affects the threat point at the bargaining stage. Notice that, by writing the transfer x as a price $\frac{1}{1+r}$ times the new balance m', net of the initial balance m, as in (14), we can re-express the left-hand side of the Euler equation as

$$u'(c)\left(q + \frac{\partial q}{\partial b'} i + \frac{1}{1+r}\frac{\partial m'}{\partial b'} + \frac{\partial \frac{1}{1+r}}{\partial b'} m'\right).$$

Importantly, the extra terms include the cross-elasticity $\frac{\partial \frac{1}{1+r}}{\partial b'}$, which mitigates the disciplining effect of spreads by making the bilateral loan cheaper as market debt increases.[16] This mitigation of the total costs of debt issuance leads the government to accumulate debt more aggressively and to delever more slowly. As in the stylized model of Section 3.2, a relational overborrowing effect arises: with discretion, the government ends up diluting legacy bondholders more than it would choose to do with commitment.

Discussion Our timing assumptions allow the government to use the issuance of market debt to influence its threat point at the bargaining stage.[17] This assumption highlights a peril of bargained bilateral loans. We believe that this timing captures the relevant interactions and that our results would also emerge under different assumptions. With the opposite timing of bilateral-then-market, the government would only be able to issue market debt after bargaining with the large lender; therefore, the threat point could include a different issuance strategy than the agreement point, reducing the scope for the cross-elasticity we emphasize, as debt issuance threats could be made off the equilibrium path. However, a timing with market-bilateral-market would recover on-path threat-point manipulation.

More generally, a model in which cash on hand could be carried across periods (for example, including reserves) would weaken the link between debt issuance and consumption, rendering the within-period timing irrelevant. In that more general model, reserves would correspond directly to cash on hand and make the government stronger. The government would be able to use debt-financed reserves to manipulate the threat point.[18] While the cross-elasticity and the relational overborrowing effect would still be present, the overall effect of the large lender's presence would then depend on the net effects of larger gross positions.

[16]This equation also emphasizes how debt issuances affect revenues from the market and from the large lender in a strictly symmetrical way (notice also that $\frac{\partial i}{\partial b'} = 1$).

[17]In the absence of a savings technology, all revenues must be consumed within the period, making any timing somewhat artificial.

[18]Bianchi and Sosa-Padilla (2024) describe how the net effect of debt and reserves goes beyond the net foreign asset position.

5.4 *Welfare effects of bilateral loans*

The forces discussed above combine to produce the welfare effects of bilateral loans. Figure 9 shows the government's value function as a function of debt b when it owes $m = 0$ to the large lender. The government prefers bilateral loans to be Unavailable during default, except of course when a default in the current period is very likely.

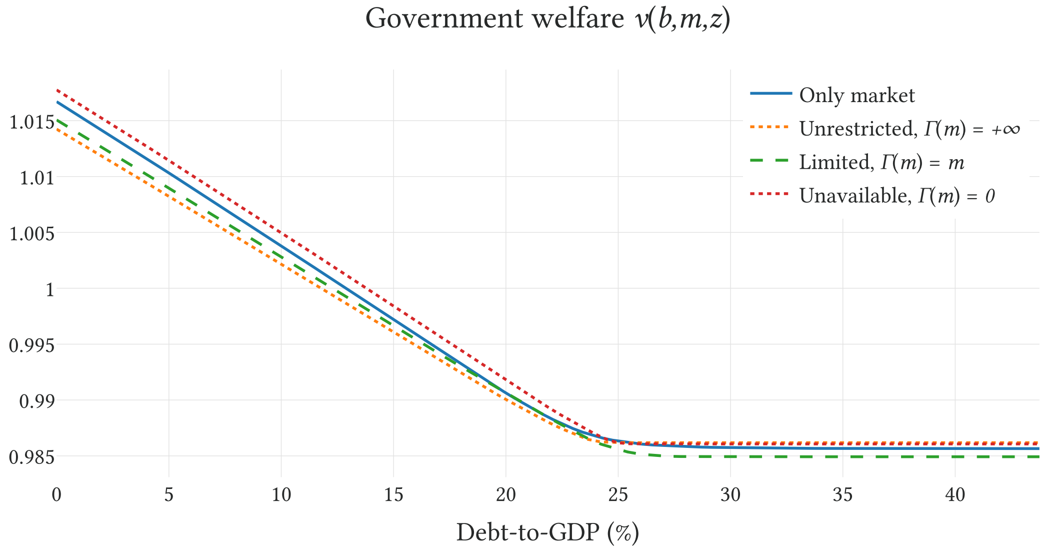

FIGURE 9: VALUE FUNCTIONS

Note: Value functions measured in consumption-equivalent terms and computed at $m = 0$ and average income, all models with bilateral loans parametrized with $\theta = 0.5$.

The presence of bilateral loans reduces the government's value at virtually all debt levels (except in the 'Unavailable' version which hardwires an extra penalty for default). A natural question is why the government might enter in such an arrangement. Figure 10 shows the welfare gains of entering an agreement with the large lender by comparing the government's value at the equilibrium with only market financing to the value at the equilibrium with bilateral loans when $m = 0$, when loans are Limited during default, conditional on repayment (the default area is shaded). Small 'islands' of welfare gains appear when income is intermediate and the government is close to the default boundary. At these states, the gains from being able to postpone default by obtaining financing from the large lender overpower the losses from higher spreads in the future equilibrium.

Figure 11 shows how welfare gains change with θ. In all cases, a government very skilled at bargaining can secure advantageous terms from the large lender, enough to extract welfare gains despite the cross-elasticity. As θ increases, such gains dissipate. The ordering of the different variants is preserved relative to the $\theta = 0.5$ benchmark: Unavailable $\succcurlyeq$ Limited $\succcurlyeq$ Unrestricted.

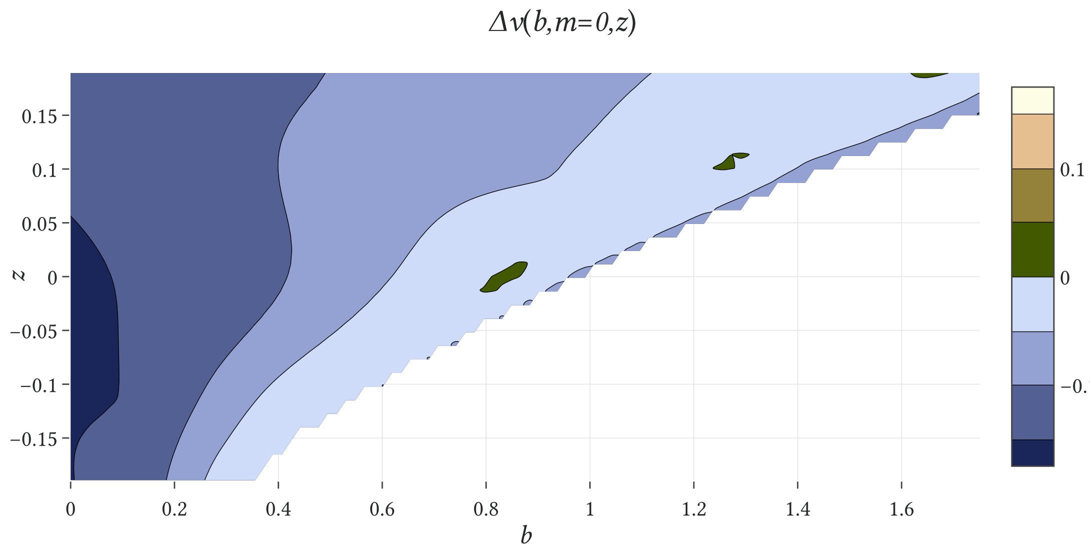

FIGURE 10: GAINS FROM ALLOWING BILATERAL LOANS (LIMITED)

Note: Welfare gains computed as the difference between the equilibrium with Limited bilateral loans ($\Gamma(m) = m$) and $\theta = 0.5$, at $m = 0$, and without bilateral loans, measured as consumption-equivalent increases.

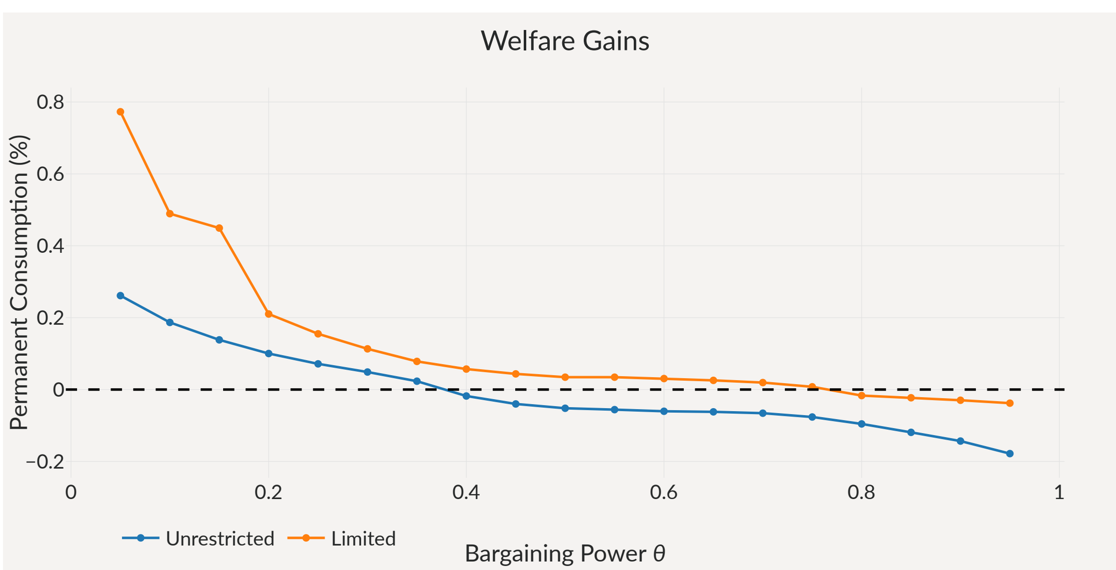

FIGURE 11: COMPARATIVE STATICS ON θ

Note: Welfare gains computed as consumption-equivalent increases averaging over the ergodic distribution of the model without bilateral loans, conditional on repayment.

6. Programming the large lender

We have shown how the combination of market debt and bargained bilateral loans generates a cross-elasticity of bilateral terms to market debt issuances, how this cross-elasticity incentivizes risk-taking and leads to a relational overborrowing effect, and how this effect heightens the frequency of default, raises spreads, and can create welfare losses for the government.

In this section we replace the bargaining protocol with fixed rules for the terms of bilateral loans, with two objectives. First, to better understand the dynamics of the relational overborrowing effect in a simpler case. Second, to explore design questions surrounding bilateral loans. Can rules that contain spreads on market debt by curbing sovereign default risk be found? Are there designs that can create welfare gains for the government? What is the optimal design of bilateral loans? Are there possible Pareto improvements, under which the government attains a higher value than without bilateral loans and also the large lender obtains profits?

To investigate these questions, we remove the bargaining protocol for bilateral loans and, instead, consider rules of the form

$$r(b', m', b) = \max\left\{r^*,\ \alpha_0 + \alpha_b(b' - (1-\delta)b) + \alpha_m m'\right\}$$

To keep the analysis focused, we conduct these exercises in the main version of the model with $\Gamma(m) = m$, corresponding to the Limited case in which bilateral loans can be refinanced during default but new drawings are forbidden. With different specifications of the Γ function, similar patterns would emerge.

With these rules, the government now faces the exogenous interest rate $r(b', m', b)$ (on the bilateral loan) when it issues $b' - (1-\delta)b$ in markets and obtains m' from the bilateral loan. Notice that because the bilateral loan always costs weakly more than the risk-free rate r^*, the large lender makes non-negative profits by construction. With this guarantee, we concentrate on outcomes for the government in what follows.

We consider three possible rules. First, a 'risk-inducing' rule with $\alpha_0 > 0, \alpha_b < 0, \alpha_m = 0$, which features the cross-elasticity found in the equilibrium with bargaining. Second, a 'size-dependent' rule with $\alpha_0 > 0, \alpha_b = 0, \alpha_m > 0$, inspired by the IMF's surcharges policy,[19] which critically does not load on market debt and therefore does not create the cross-elasticity. Third, an optimally designed rule that optimizes the government's welfare over the parameters $(\alpha_0, \alpha_b, \alpha_m)$. Table 5 summarizes our findings.[20]

[19] The cost of IMF programs increases with the size of the program via surcharges which apply after the amounts borrowed exceed certain thresholds. There are also surcharges that apply as a function of time, which we do not capture.

[20] The size-dependent rule is parametrized with $(\alpha_0, \alpha_b, \alpha_m) = (0.01, 0, 0.5)$, while the risk-inducing rule uses $(\alpha_0, \alpha_b, \alpha_m) = (0.085, -2.875, 0.05)$, and the optimal rule uses $(\alpha_0, \alpha_b, \alpha_m) = (0, 0.25, 0.125)$.

Table 5: Outcomes with exogenous rules for bilateral loans

	Only market	**Size dependent** r	**Risk inducing** r	**Optimal** r	**Limited,** $\Gamma(m) = m$
Avg spread (bps)	630	413	831	227	1,209
Std spread (bps)	474	240	447	130	860
$\sigma(c)/\sigma(y)$ (%)	104	105	104	106	100
Debt to GDP (%)	16.4	17.3	17.1	17.7	17.6
Loan to GDP (%)	0	0.727	0.486	2.56	2.22
Loan spread (bps)	–	662	296	516	-126
Corr. loan & spreads (%)	–	49.4	-29.2	38.9	67.7
Default frequency (%)	5.03	3.2	6.19	1.92	9.04
Welfare gains (rep)	–	0.15%	-0.097%	0.69%	-0.038%

Note: Statistics are based on simulations of the ergodic distribution of each model, conditional on repayment. 'Only market' refers to a version of the model without bilateral loans, while 'Limited' refers to the version in which bilateral loan balances cannot be increased during default (with $\theta = 0.5$). The middle columns denote the versions with exogenous terms for the bilateral loan: 'Size dependent r' has the bilateral interest rate increase with the balance in the bilateral loan, while 'Risk inducing r' has the bilateral interest rate decrease with the amount of market debt. Default frequencies are computed on the ergodic distribution. Welfare gains are given as equivalent increases in consumption calculated over the ergodic distribution of the model without bilateral loans, conditional on repayment.

The risk-inducing rule creates similar dynamics to the equilibrium with bargaining and, thus, welfare losses for the government. However, the size-dependent rule exemplifies the possibility of rules which induce lower default and spreads, improving the government's welfare, while also generating profits for the large lender.

The rule that optimizes the government's welfare features $\alpha_b > 0$ and hence embodies the opposite cross-elasticity as the risk-inducing rule and the equilibrium with bargaining. The optimal rule curbs the government's incentives to accumulate debt, through the same logic that prevails in the equilibrium with bargaining, only reversed. By increasing the cost of entering or remaining in the risky region of the state space, the optimal rule helps the government to avoid default but also helps it mitigate the debt dilution problem, raising prices for market debt. This benefits the government even when debt is low and default risk is contained.

7. Concluding remarks

Inspired by the rise of new bilateral lenders outside the Paris Club and with claims to de facto seniority relative to private creditors, we investigate the interaction between marketable debt and bilateral debt. We develop a model of sovereign debt in which the government can borrow from a large lender as well as from a competitive fringe of private creditors. We find that both sources of funds are linked by strong interactions, even when quantitatively the amounts borrowed from the large lender remain an order of magnitude smaller than market debts.

The model features a 'relational overborrowing' effect, which arises in the presence of the large lender. This effect is due to an endogenous cross-elasticity of bilateral borrowing terms to outcomes in debt markets, which erodes the market discipline of spreads. Our model of bargaining with the large lender generates this cross-elasticity with three key ingredients: loans from the large lender are more difficult (or impossible) to default, their terms result from bargaining, and they are of shorter maturity than marketable debts. These ingredients capture the defining features of Central Bank swap lines, which Horn et al. (2021b) identify as a key instrument of senior bilateral sovereign lending. The simpler version of the model with exogenous bilateral terms shows that the cross-elasticity is in itself sufficient to incentivize the government to overborrow, face more sovereign risk, and ultimately attain lower levels of welfare.

The model with exogenous bilateral terms also illustrates a broader point. Policy discussions surrounding bilateral debts often focus on the price of the loans and on seniority itself. Our results emphasize that welfare losses are likely to come from a different source. In particular, we show how welfare losses can arise even when the government is free not to borrow from the bilateral lender if the terms offered are not advantageous enough relative to the market. We also provide examples of pricing schedules for the senior debt, like size-dependent rules, which improve welfare for the government without losses for the large lender. Rather, our main result is that the perverse incentives created by the cross-elasticity, which have been more overlooked, are crucial to generating the welfare losses we find. Our main model then describes a set of plausible assumptions, especially sequential bargaining over borrowing terms, under which the cross-elasticity could be expected to arise, thus creating perils of bilateral sovereign debt.

Our results suggest that increasing the number of financing sources is not necessarily beneficial for the borrowing government. While bilateral loans can help a government fend off default, they can also make it more likely: either through reducing the effective costs of default or through the relational overborrowing effect.

The welfare impact of the large lender's presence raises important policy questions and challenges. Limiting the use of bilateral debt during defaults is a clear welfare-enhancing policy in this model. Because the relational overborrowing effect operates through increased default risk, the gains of fiscal rules that constrain market borrowing should be larger for countries with access to the type of bilateral debts we describe.

The model implies a simple test to gauge the likely effects of a new bilateral creditor or instrument in practice. Bilateral loans whose interest rate is expected to be strongly decreasing in the amount (or spreads) of marketable debt will induce relational overborrowing and are thus likely to hurt welfare. More generally, it highlights the benefits of transparent rules for the terms of bilateral debts.

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A. More Results

Increasing the borrower's bargaining power Figure 12 shows that when the borrower holds all the bargaining power, the loan interest rate is constant at β_L^{-1}. In this example with $\beta = \beta_L$, the upper bound on how large the loan m does not bind, but in the quantitative version (where market and bilateral debt coexist) with $\beta < \beta_L$, which at $\theta = 0$ recovers the model of Hatchondo, Martinez, and Önder (2017), the borrower prioritizes bilateral loans as a source of funding and thus quickly reaches the upper bound.

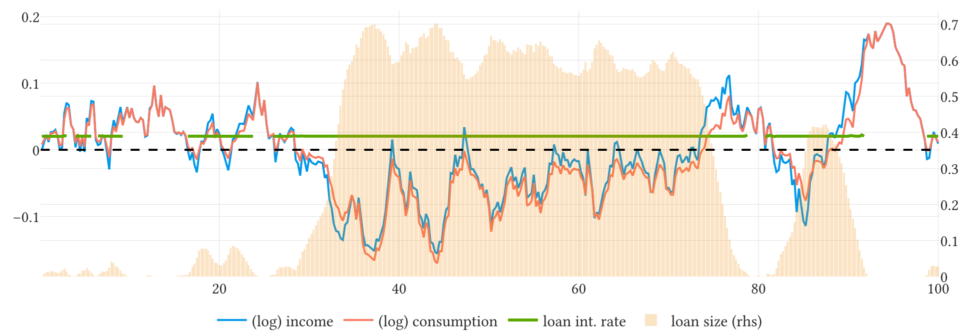

Figure 12: Simulated path, $\theta = 0$

Dynamics of loans around defaults Figure 13 shows that, conditioning on an exclusion period of 2 years, the economy issues debt in the market in order to pay off the loan as soon as it recovers market access.

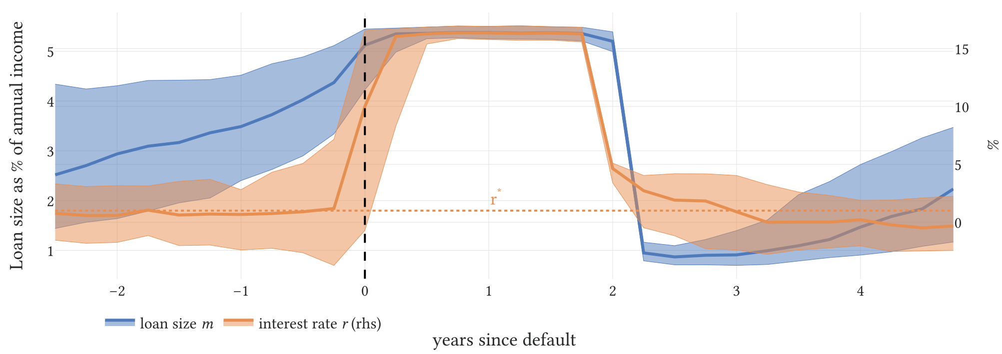

Figure 13: Loans around default events

Note: Event studies around default of market debt, with Unrestricted ($\Gamma(m) = +\infty$) bilateral loans and $\theta = 0.5$, conditional on the default spell lasting exactly two years.